



Computational Social Science from Space

Ingmar Weber

Sep 18, 2025

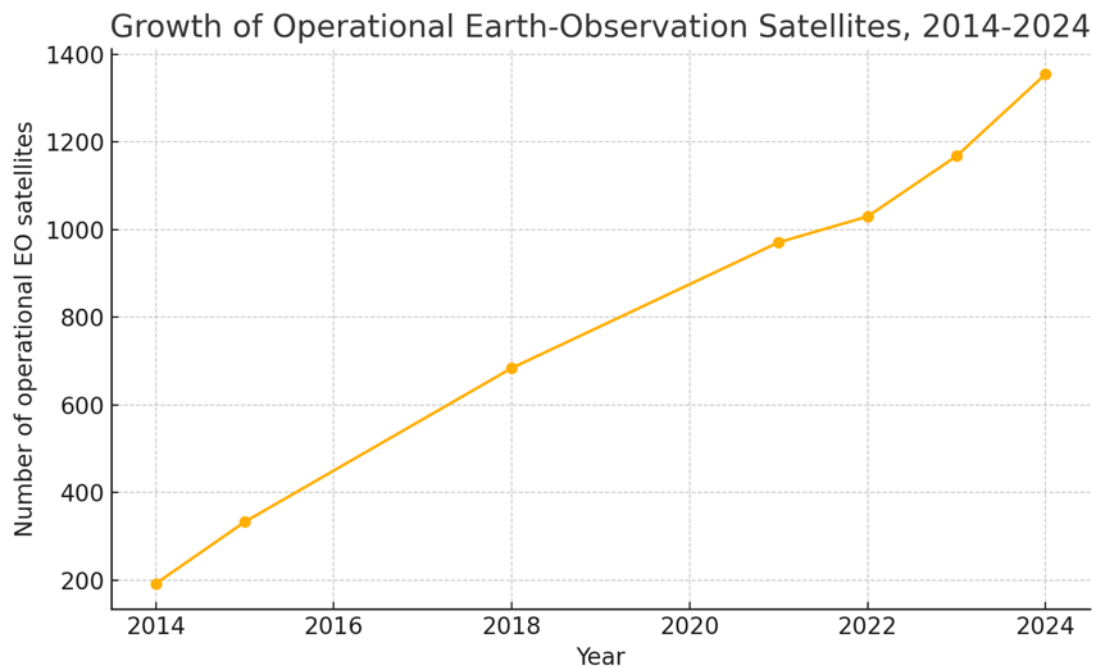
A man in a black tuxedo and bow tie is seated at a dark podium outdoors. On the podium, from left to right, are a bottle of water, a glass, a microphone, and a white rotary telephone. The background is a blurred field of tall grass. The text "AND NOW FOR SOMETHING COMPLETELY DIFFERENT." is overlaid in white, bold, sans-serif font at the bottom of the image.

**AND NOW FOR SOMETHING
COMPLETELY DIFFERENT.**

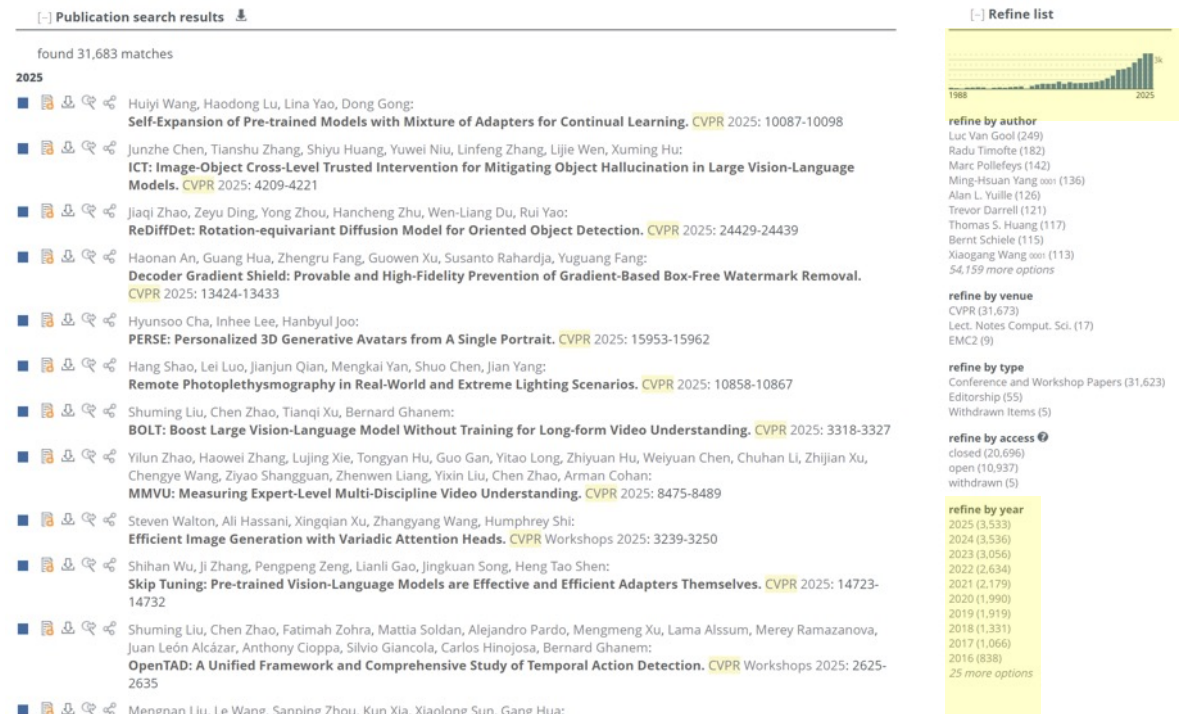
What if you had access to CCTV data for any place on earth going back several decades?



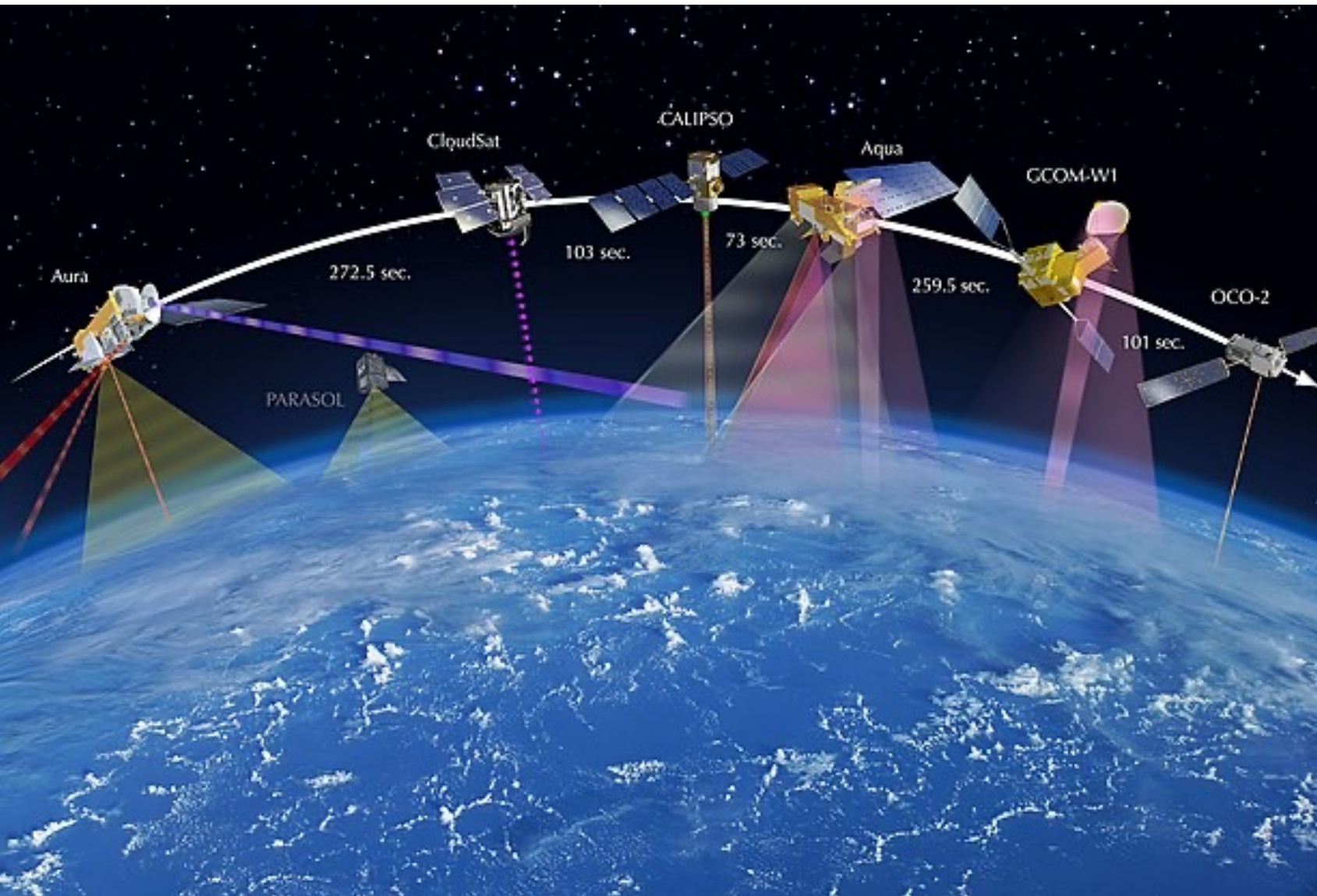
Rapidly Increasing Capabilities



+



Earth Observation Satellites (EO)

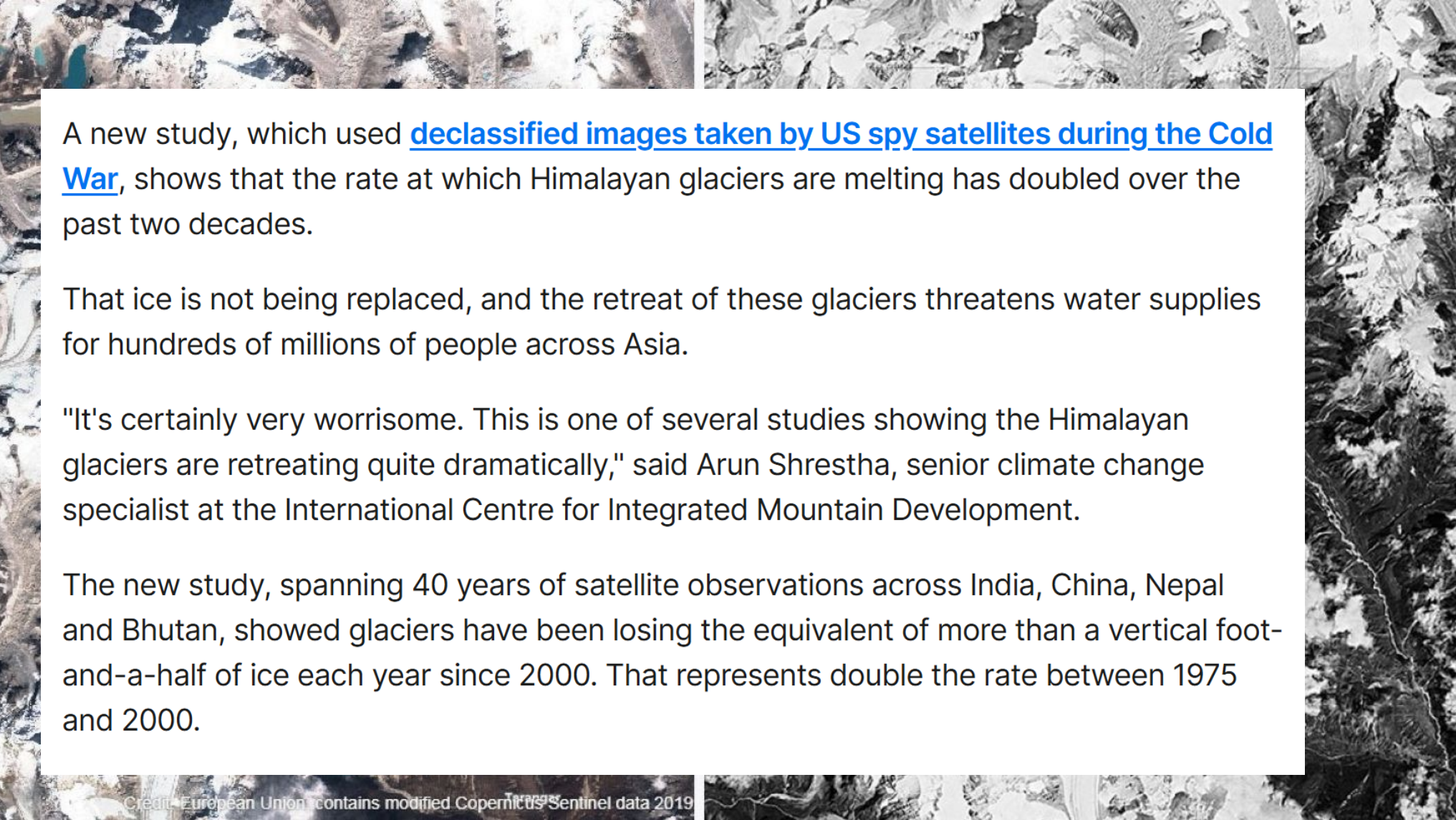


- > 1000 EO satellites
- 30cm - 20m resolution
- High-res is expensive
- Low-res is free
- Daily to monthly updates
- Optical, infrared, radar, ...
- Archival data ~20 years

What Can You See from Space?

And what is relevant to CSS?

Melting Glaciers



A new study, which used [declassified images taken by US spy satellites during the Cold War](#), shows that the rate at which Himalayan glaciers are melting has doubled over the past two decades.

That ice is not being replaced, and the retreat of these glaciers threatens water supplies for hundreds of millions of people across Asia.

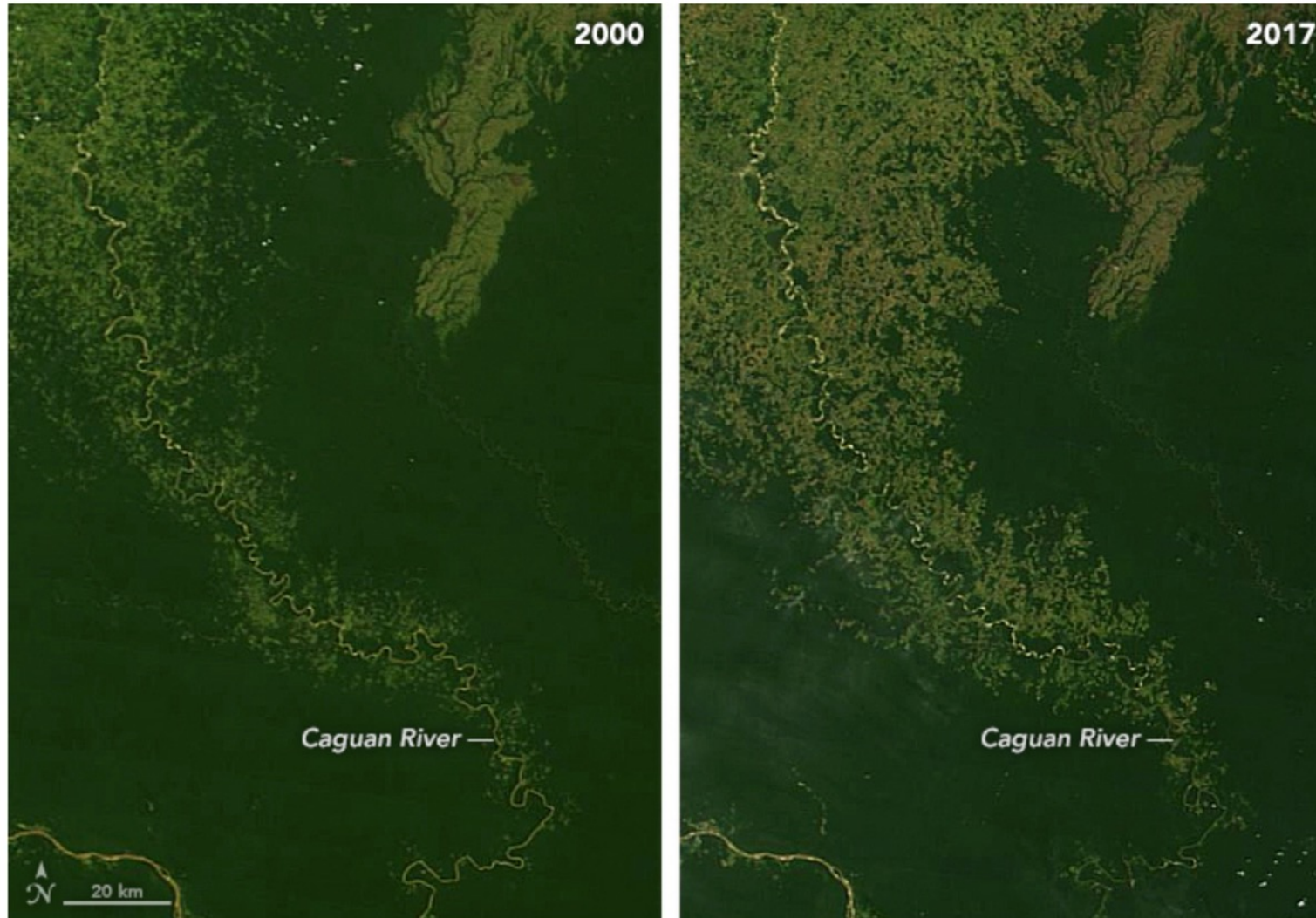
"It's certainly very worrisome. This is one of several studies showing the Himalayan glaciers are retreating quite dramatically," said Arun Shrestha, senior climate change specialist at the International Centre for Integrated Mountain Development.

The new study, spanning 40 years of satellite observations across India, China, Nepal and Bhutan, showed glaciers have been losing the equivalent of more than a vertical foot-and-a-half of ice each year since 2000. That represents double the rate between 1975 and 2000.

Credit: European Union contains modified Copernicus Sentinel data 2019

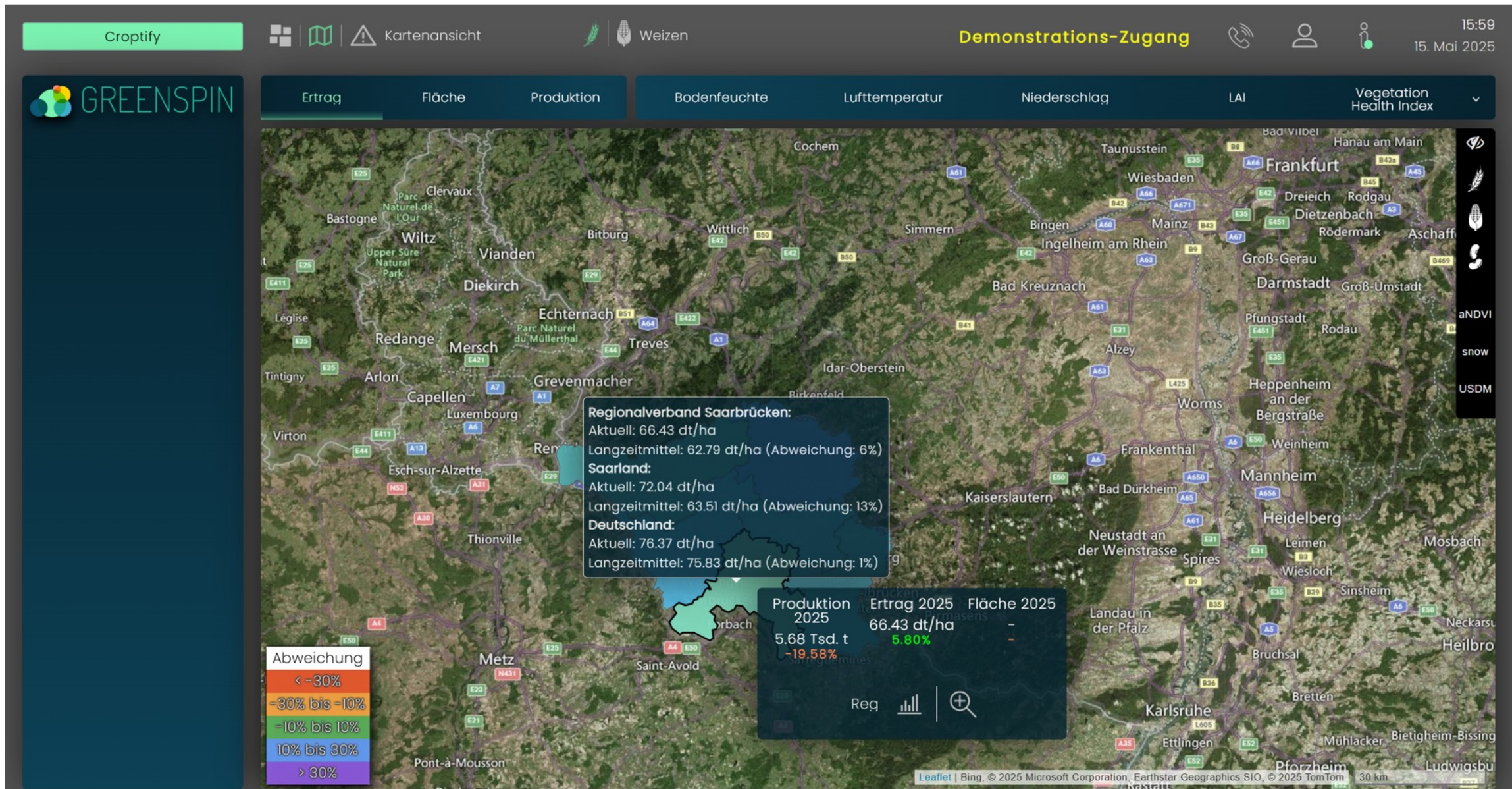
<https://www.euronews.com/2019/06/20/himalayan-glaciers-melting-at-alarming-rate-satellite-data-shows>

Deforestation in the Amazon

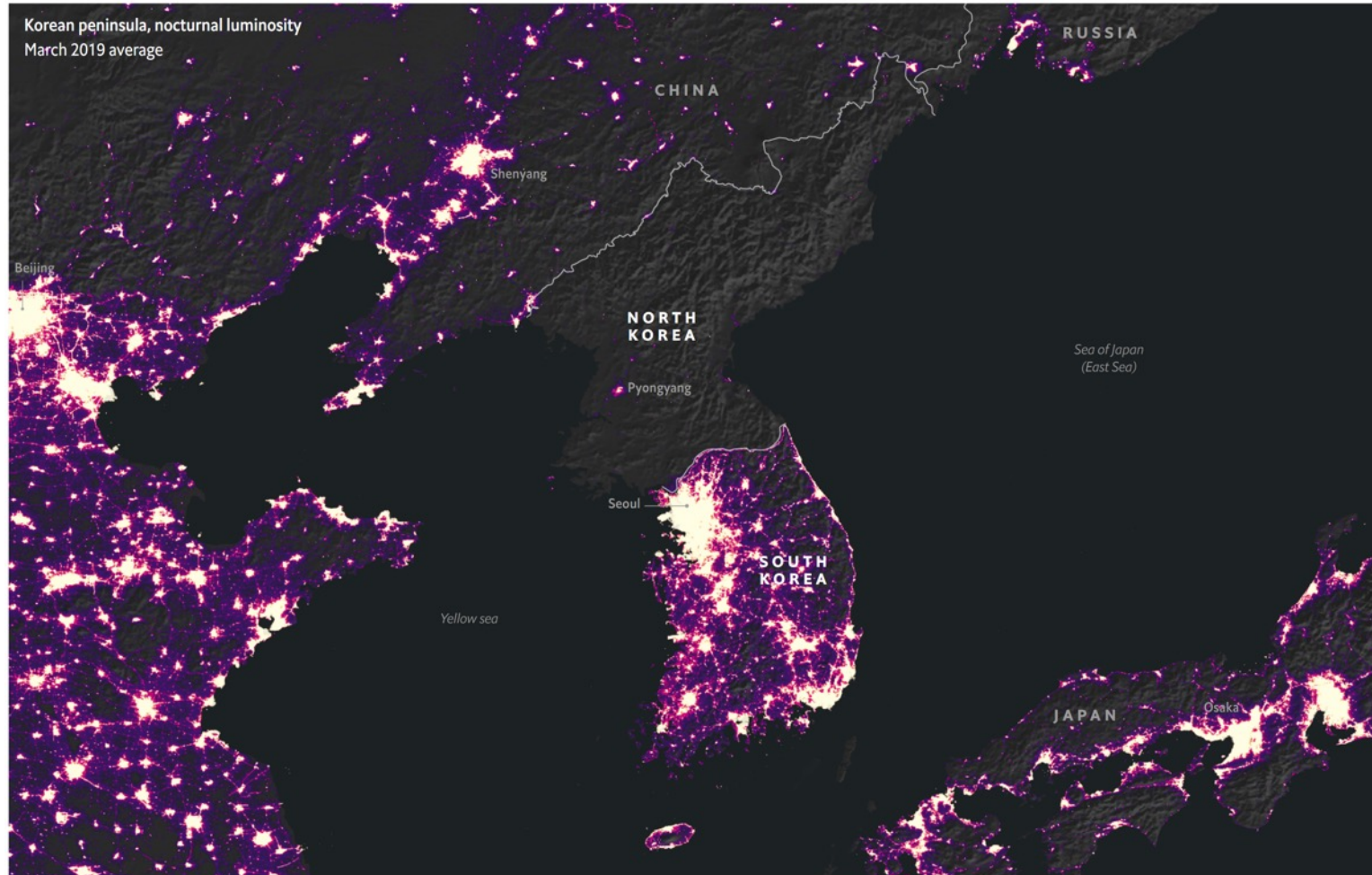


<https://earthobservatory.nasa.gov/images/145888/making-sense-of-amazon-deforestation-patterns>

Crop Yield Monitoring and Prediction



Night Light in North Korea



<https://www.economist.com/graphic-detail/2019/05/04/satellite-data-shed-new-light-on-north-koreas-opaque-economy>

Night Light in Syria 2012



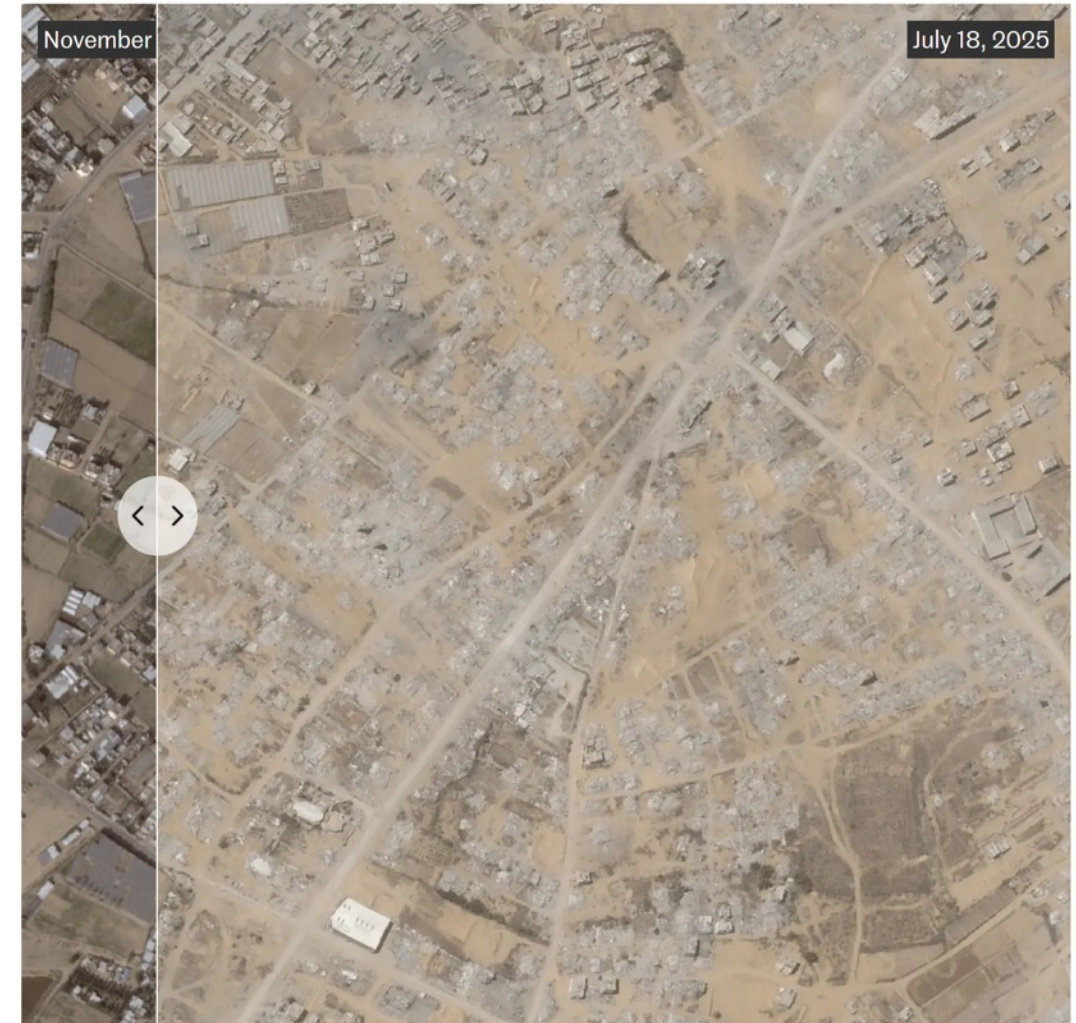
<https://earthobservatory.nasa.gov/images/90100/night-lights-change-in-the-middle-east>

Night Light in Syria 2016



<https://earthobservatory.nasa.gov/images/90100/night-lights-change-in-the-middle-east>

Destruction in the Gaza War



Satellite images of the south of Khan Yunis, in the Gaza Strip. PLANET LABS

https://www.lemonde.fr/en/international/article/2025/07/28/gazan-city-of-khan-yunis-is-almost-completely-leveled-satellite-images/show_6743817_4.html

Mapping Wealth and Poverty



Detecting Tax Evasion from Space

SPiEGEL International

Finding Swimming Pools with Google Earth

Greek Government Hauls in Billions in Back Taxes

In a bid to increase revenues, the Greek authorities are employing all kinds of clever tricks to crack down on tax cheats, including using Google Earth to find undeclared swimming pools. But efforts by the government to liberalize markets could unleash a wave of civil unrest.

Von **Daniel Steinvorth**
02.08.2010, 14.48 Uhr

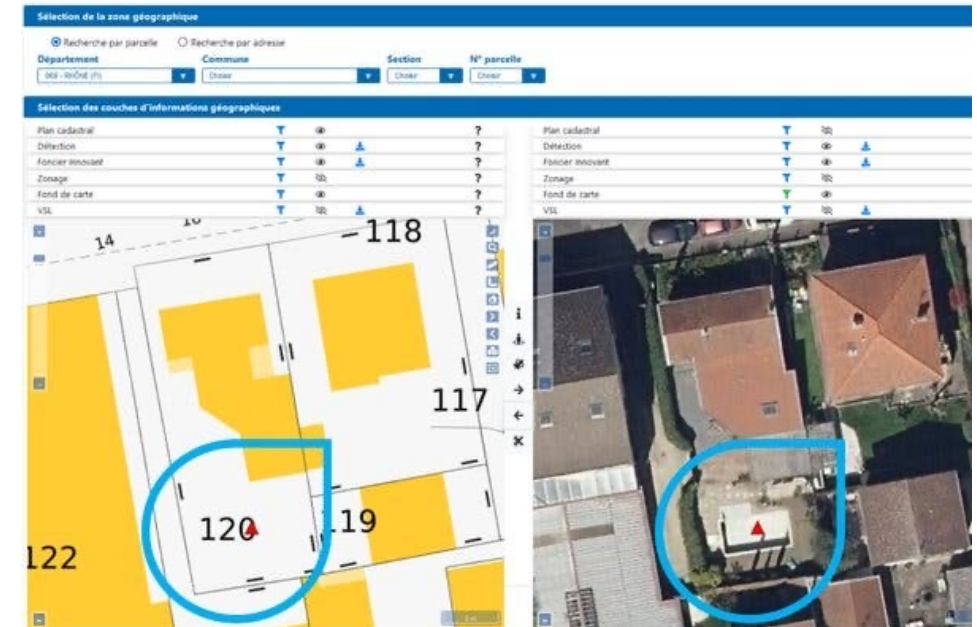
“They use satellite pictures by Google Earth to locate country villas, swimming pools and properties. And these tactics have revealed that the suburbs didn't have 324 swimming pools, as was reported, but rather 16,974.”

<https://www.spiegel.de/international/europe/finding-swimming-pools-with-google-earth-greek-government-hauls-in-billions-in-back-taxes-a-709703.html>

The New York Times

French Tax Collectors Use A.I. to Spot Thousands of Undeclared Pools

Algorithms combing through satellite photos found over 20,000 unreported swimming pools in a few regions, yielding an expected \$10 million in taxes, and the system will soon go nationwide.



<https://www.nytimes.com/2022/08/30/world/europe/france-taxes-pools-artificial-intelligence.html>

Racism and Climate Impact

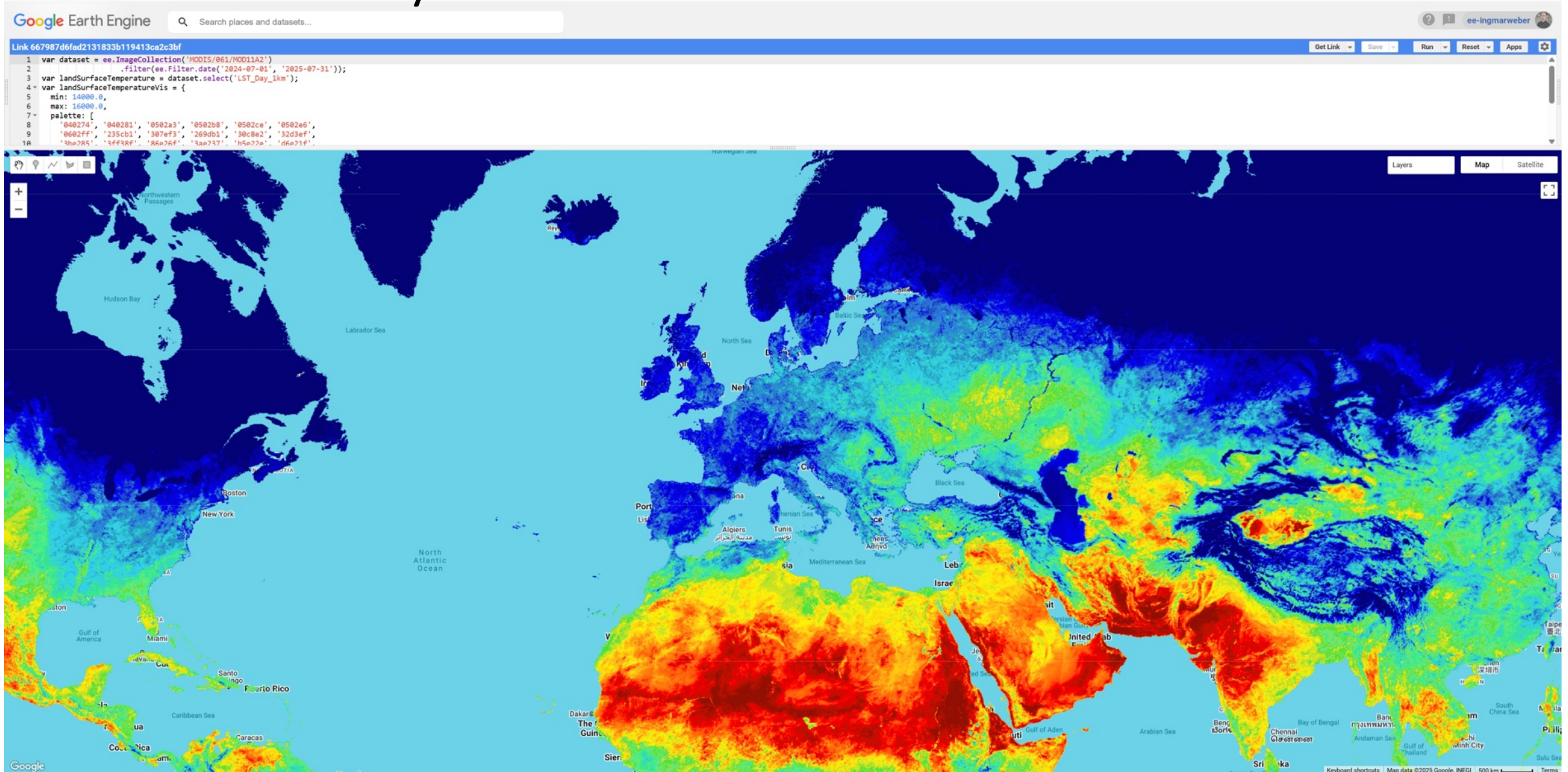


In cities like Baltimore, Dallas, Denver, Miami, Portland and New York, neighborhoods that are poorer and have more residents of color can be 5°-20° Fahrenheit (3°-5°C) hotter in summer than wealthier, whiter parts of the same city.



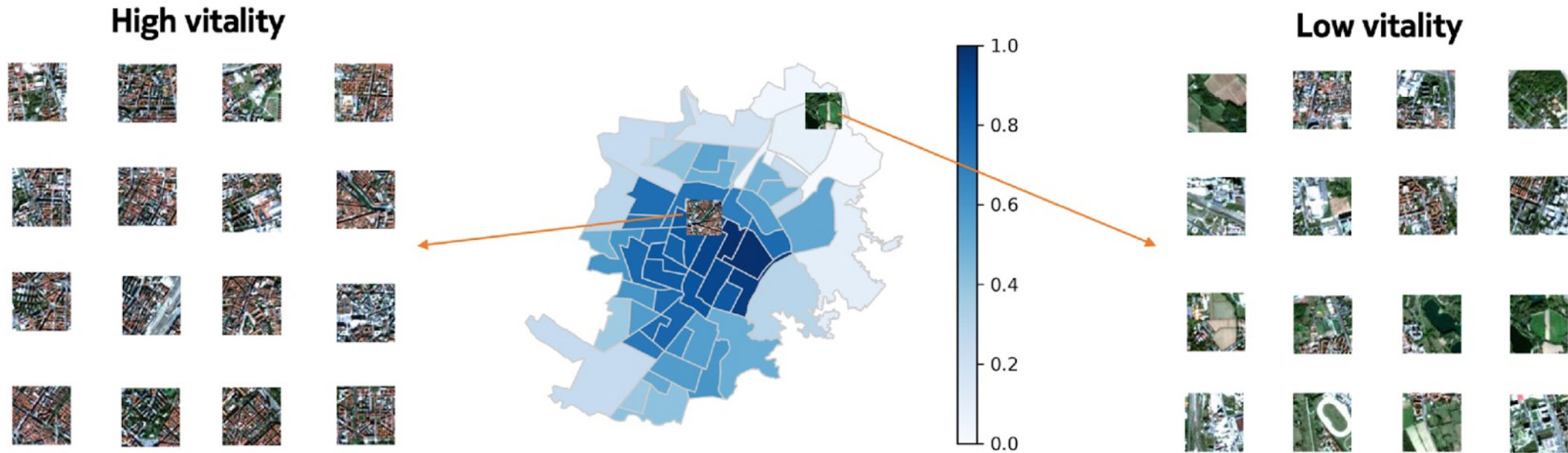
<https://www.nytimes.com/interactive/2020/08/24/climate/racism-redlining-cities-global-warming.html>

Easily Accessible Data via GEE



<https://code.earthengine.google.com/667987d6fad2131833b119413ca2c3bf>

Urbanism



“Back in the 1970s, Jane Jacobs theorized urban vitality and found that there are **four conditions** required for the **promotion of life in cities: diversity of land use, small block sizes, the mix of economic activities, and concentration of people.** [...] Here we propose the use of one single source of data, which happens to be publicly available: Sentinel-2 satellite imagery. In particular, since the first two conditions (**diversity of land use and small block sizes**) are **visible to the naked eye from satellite imagery**, we tested whether we could automatically extract them with a state-of-the-art deep-learning framework and whether, in the end, the extracted features could predict vitality.”

Jane Jacobs in the Sky: Predicting Urban Vitality with Open Satellite Data

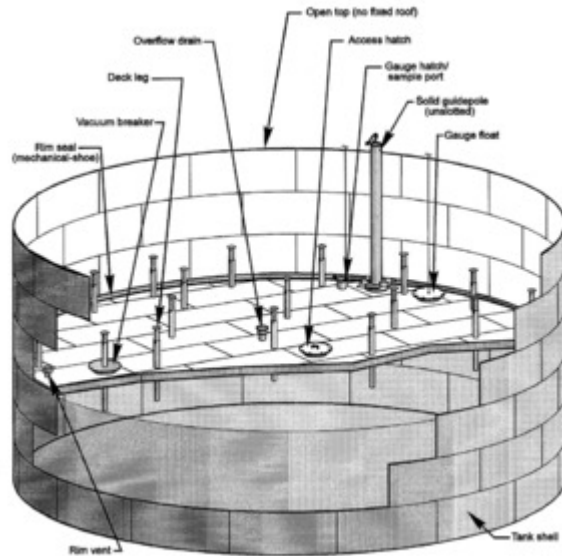
<https://dl.acm.org/doi/10.1145/3449257>

Voting Patterns from Space

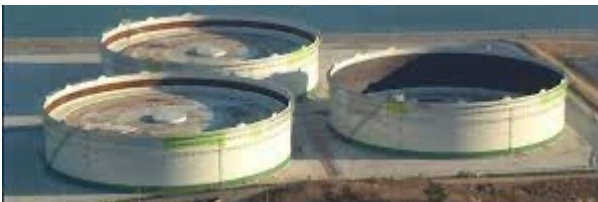


Representation of Politics from Space, by Vedran Sekara,
<https://vedransekara.github.io/>

Floating Roof Oil Tanks



https://en.wikipedia.org/wiki/External_floating_roof_tank



<https://www.youtube.com/watch?v=aKlpZU95jco>



<https://medium.com/from-the-macroscope/as-world-panics-over-oil-storage-orbital-data-shows-2b-barrels-of-space-left-d6745216e946>

Economic Activity



(a) Airport before and after the COVID-19



(b) Car rental parking lot before and after the COVID-19

Epidemics?

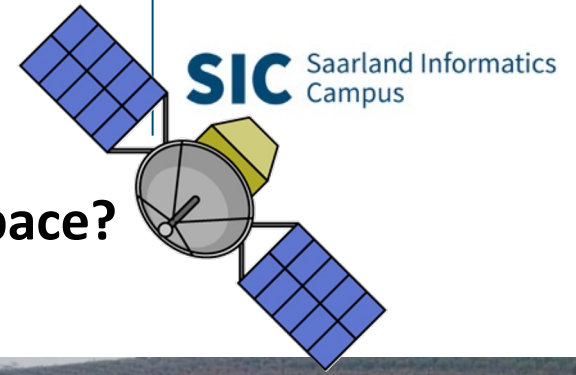


Examples of Our Work



Displacement in Syria

Can we see this from space?

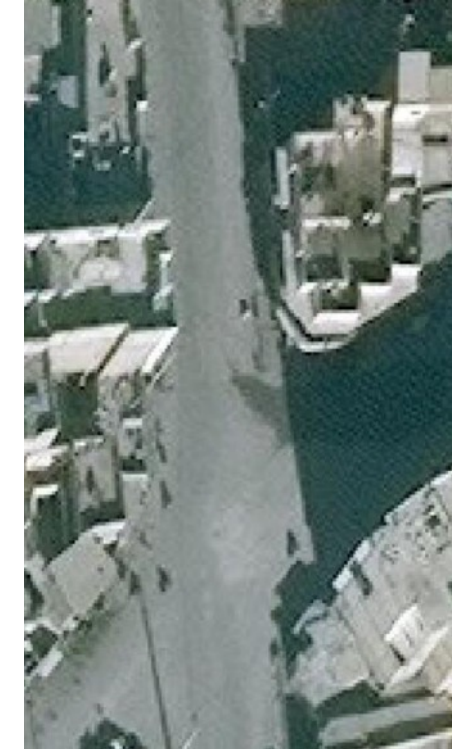


The Basic Idea: Two Variants



Vehicle Flows

Look for cars going from A to B
Captures movement in the moment



Vehicle Stocks

Two snapshots: difference in vehicle counts
Captures before-after differences

SecureWatch

36.36124,37.5023

Map Options

My Imagery

Use With

Usage

Cart

☐ Explore Mode

SecureWatch 46066360

Ingmar Weber

?

☒ Google Maps

☐ OpenStreetMap

Map Transparency

LAT: 36.37282 : LON: 37.53809

SRTM2 - 30m: 442 m. 1450 ft

☰



Filters

×

- Acquisition Date (None) ☐
- Cloud Cover 10% ☒
- Source (None) ☐
- Sensor Type (None) ☐
- Content Type (None) ☐
- Max GSD 50cm ☒
- Image Bands (None) ☐
- Off Nadir Angle (None) ☐
- Sun Elevation (None) ☐
- Target Azimuth (None) ☐

Close

Reset

Apply

»

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2017-02-27

2016-12-08

2016-11-27

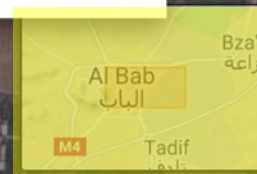
2016-11-27

2016-11-25

2016-11-22

2016-11-22

2016-11-19



Using Existing Source Code

←

→

↺

🏠

🔒 https://github.com/maups/covid19-satellite-analysis

⋮

🔒

📄

🌟

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💬

maups / covid19-satellite-analysis

👁 Watch 2

☆ Star 9

🍴 Fork 3

<> Code

! Issues

🔗 Pull requests

▶ Actions

📁 Projects

📖 Wiki

🛡 Security

📈 Insights

🔗 master 1 branch 0 tags

Go to file

Add file

📄 Code

🗨

Maurício Pamplona Segundo Update images and add annotations

ae03187 on Jul 14 10 commits

📁 assets	Delete report.pdf	6 months ago
📁 images	Update images and add annotations	3 months ago
📄 LICENSE	Update LICENSE	6 months ago
📄 README.md	Update README.md	6 months ago
📄 download_models.sh	Source and scripts for object detection	6 months ago
📄 inference.py	Source and scripts for object detection	6 months ago
📄 parameters.py	Source and scripts for object detection	6 months ago
📄 requirements.txt	Source and scripts for object detection	6 months ago
📄 run_inference.sh	Source and scripts for object detection	6 months ago
📄 show_regions.py	Source and scripts for object detection	6 months ago

README.md

About

No description, website, or topics provided.

📖 Readme

📄 View license

Releases

No releases published

Packages

No packages published

Languages

Python 94.2%

Shell 5.8%

Output for Al Bab, Syria



Manual Validation



56 manually identified cars



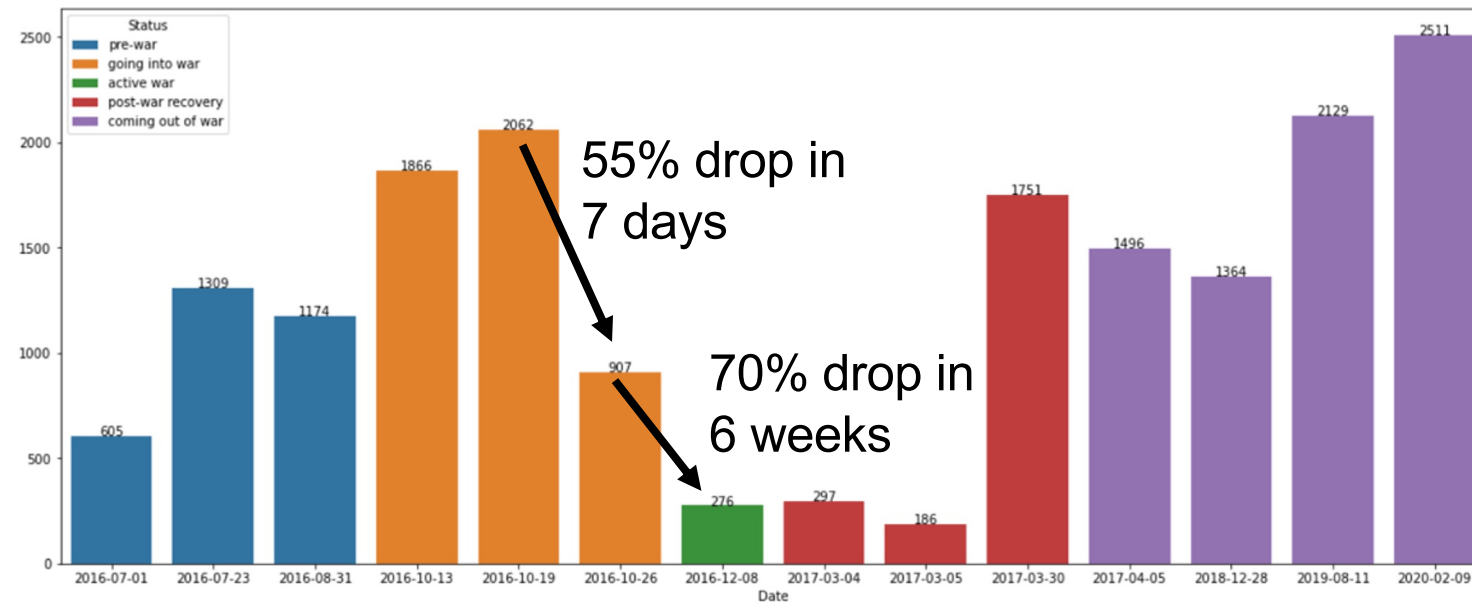
48 automatically identified cars

Recall: $46/56 = 82\%$

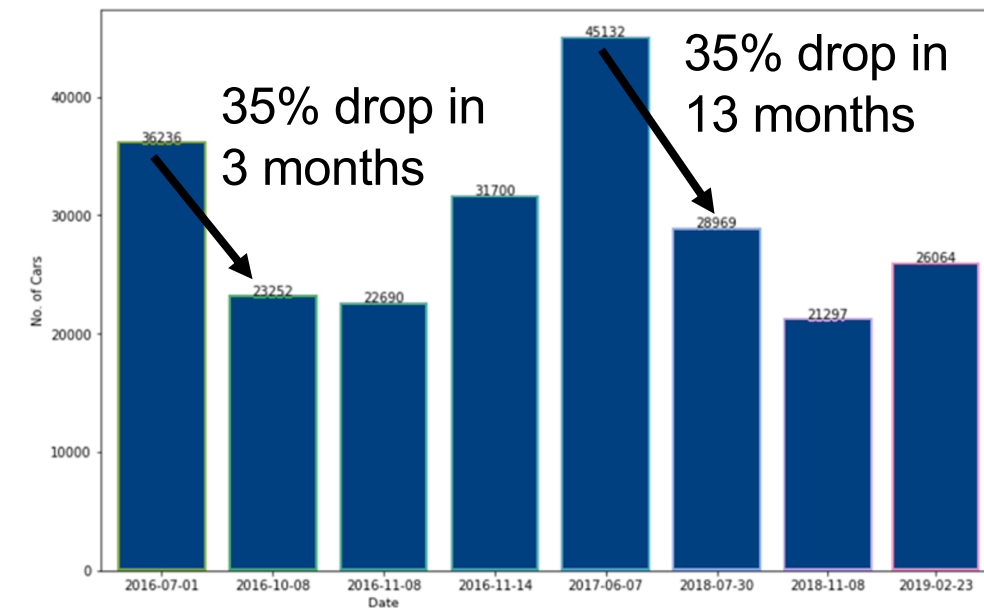
Precision: $46/48 = 96\%$

In clear, cloud-free images ...

Example Results



Al Bab



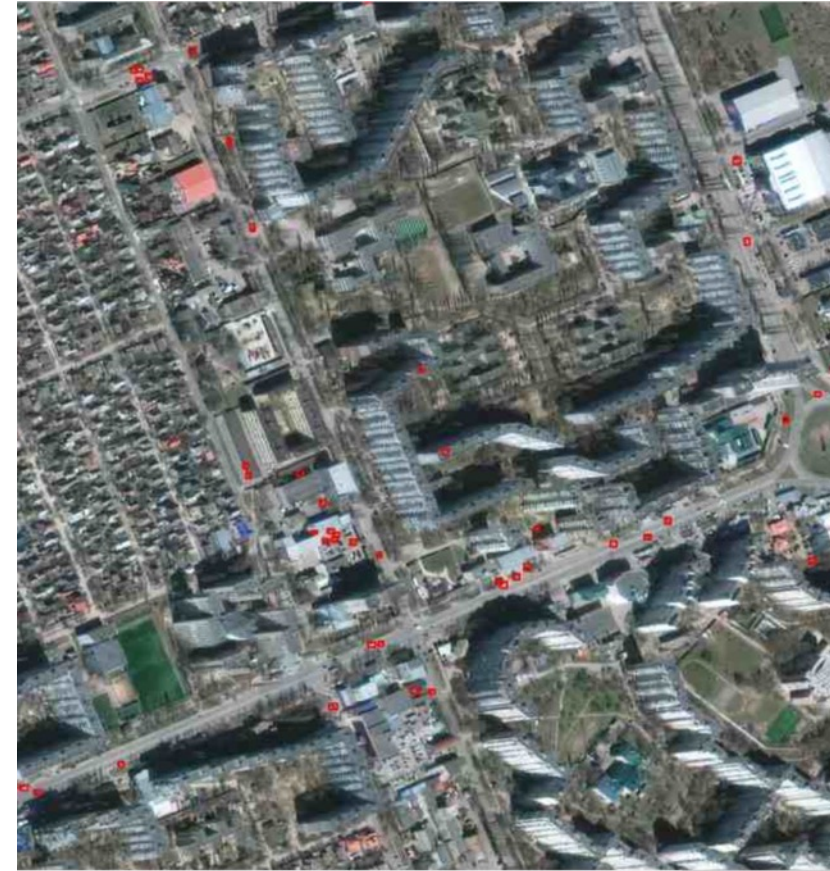
Latakia

Output for Kherson, Ukraine

Before invasion

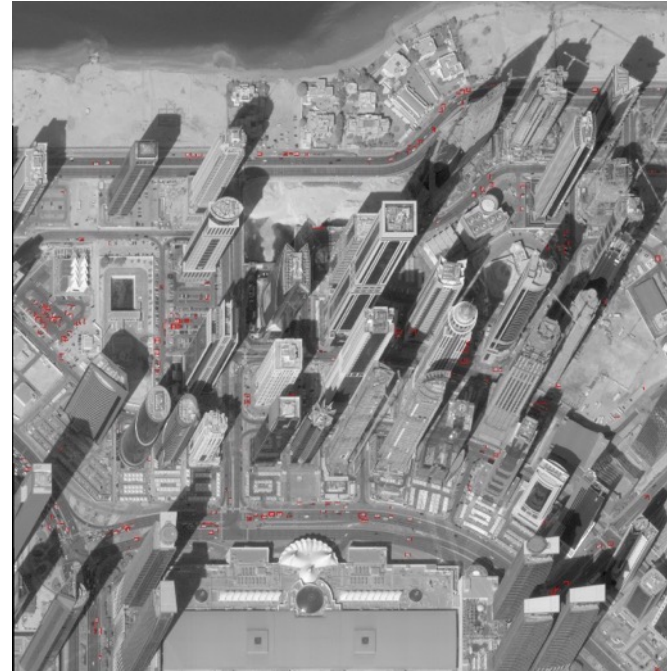
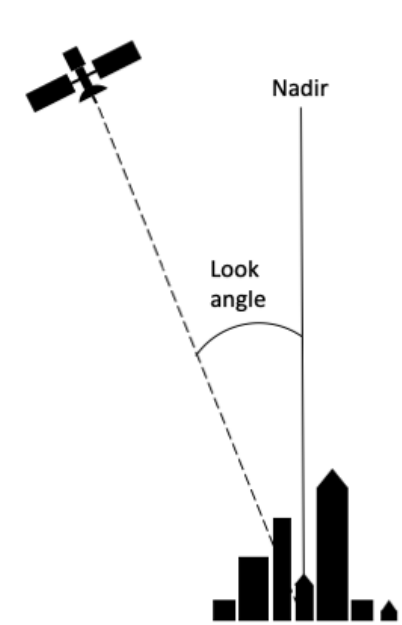


After invasion



Different lighting, different nadir angle, different day of week, ...

Example Challenge: Nadir



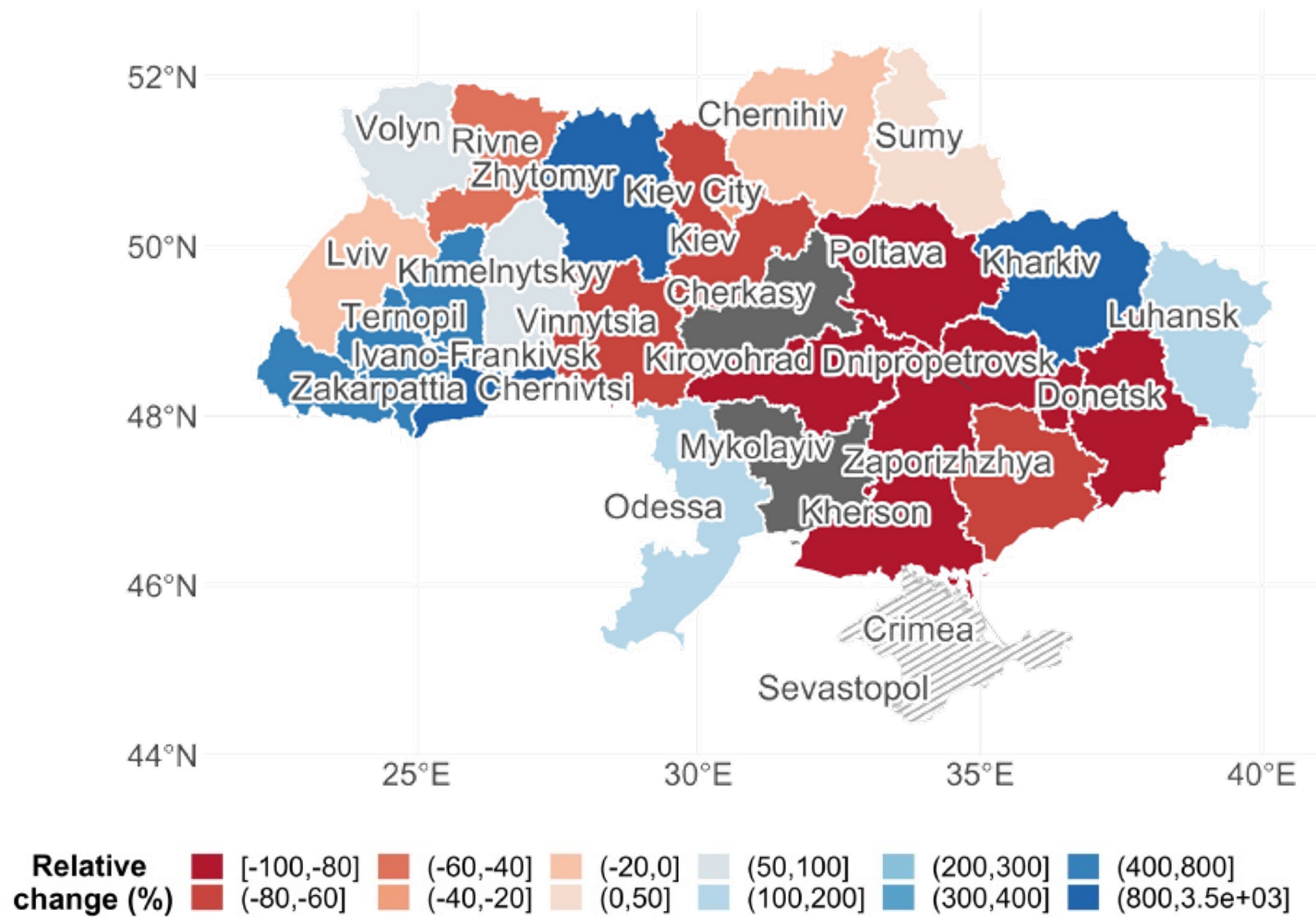
918 cars detected



3827 cars detected

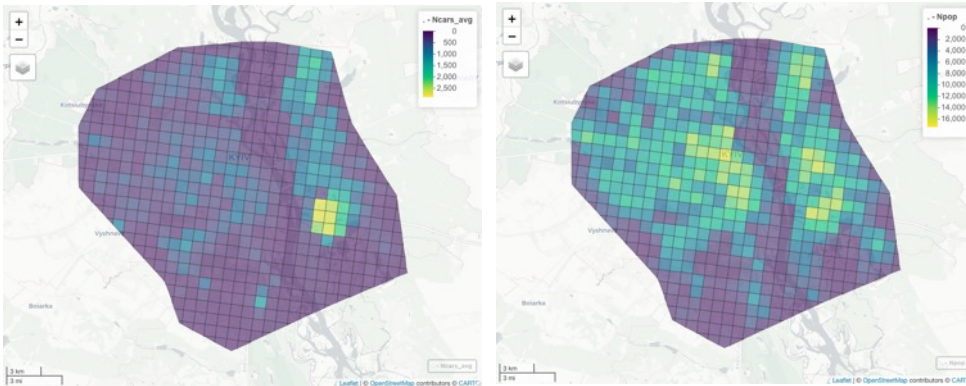
<https://medium.com/the-downling/challenges-with-spacenet-4-off-nadir-satellite-imagery-look-angle-and-target-azimuth-angle-2402bc4c3cf6>

Car Density 2019 -> Post-Invasion 2022



From Cars to Population Estimates

Historic data for calibration
cars vs. gridded population



Pre-War # Cars

Pre-War Population

Global

Learn a *city-wide* correspondence:

Any grid cell in Kiev with x cars corresponds to y people

+ Can learn that $x=0$, but $y \neq 0$

+ Can fit any functional shape

- Fooled by local anomalies, such as industrial areas

Local

Learn a *per-grid-cell* correspondence

For grid cell i , x_i cars corresponds to y_i people

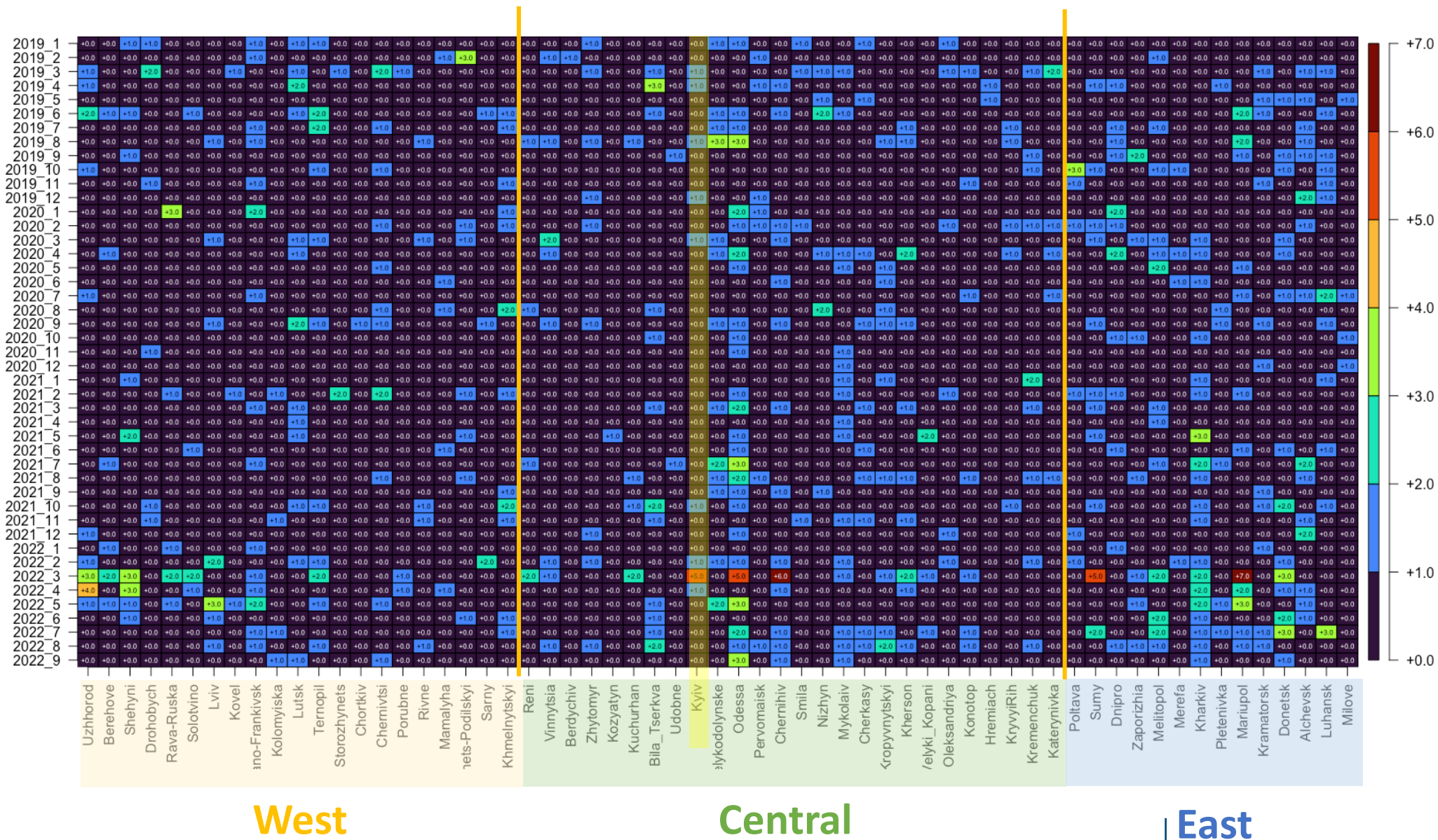
+ Naturally adapts to local anomalies

- Assumes a $x_i/2 \Rightarrow y_i/2$

Very simple yet effective. Could incorporate many more covariates:

- Public transportation, roads, population density, (street) parking availability, ...

MAIAR



Can we Use *Daily* Satellite Images?



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Daily Earth Data to See Change and Make Better Decisions

Planet provides daily satellite data that helps businesses, governments, researchers, and journalists understand the physical world and take action.

< 1 / 9 >

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NEWS

Planet Completes Acquisition of Salo Sciences

With the purchase of Salo Sciences, Planet plans to further develop its offerings to enable customers to globally verify and monitor forest change, quantify carbon stocks, and mitigate climate risks.

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<https://www.planet.com/markets/education-and-research/>

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September 16, 2025
PlanetScope Scene
3m/px 100%
2 ITEMS

September 15, 2025
PlanetScope Scene
3m/px 100%
2 ITEMS

September 14, 2025
PlanetScope Scene
3m/px 100%
2 ITEMS

September 13, 2025
PlanetScope Scene
3m/px 100%
1 ITEM

September 11, 2025
PlanetScope Scene
3m/px 100%
2 ITEMS

September 10, 2025
PlanetScope Scene
3m/px 100%
2 ITEMS

September 6, 2025
PlanetScope Scene
3m/px 100%
3 ITEMS

September 5, 2025

API (:) ORDER (0)

Europa-Institut, Saarland, Germany

Mapbox OpenStreetMap

49.25336° N, 7.04716° E 49.25315° N, 7.04548° E 17.11 0.723 m/px 50 m

Sep 2025

How can you count cars, without seeing cars?

Day of the Week as Weak Annotation



The model learns, to estimate car park occupancy.

Given a pair of images, which one is fuller?

Parking lot size	AUC
large parking lots	0.92
medium parking lots	0.91
small parking lots	0.65

Performance on 20% of unseen parking lots

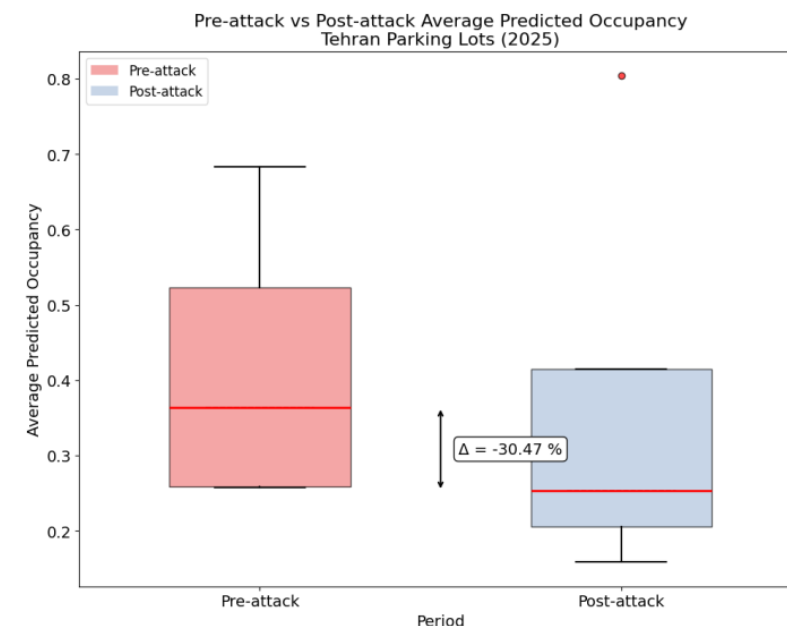


Image 1

Same parking lot



Image 2

Evolution of Highway Speeds



Risks

Abuse Risk for High-Res Imagery

ThePrint



Home > Defence > Controversial Pakistani firm BSI removed as partner from US satellite company's website

Defence India

Controversial Pakistani firm BSI removed as partner from US satellite company's website

Hours after ThePrint revealed that Maxar had been receiving orders for high-resolution satellite images of Pahalgam, including 10 days before the attack, the company removed the Pakistani firm from its website.

SOUMYA PILLAI 10 May, 2025 04:23 pm IST



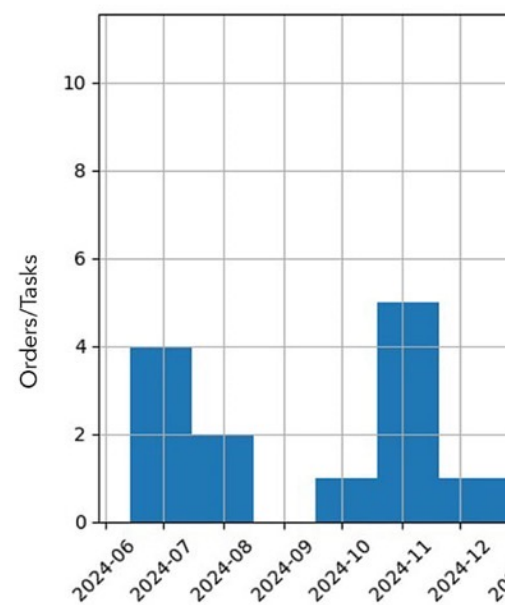
The home page of Maxar Technologies | Photo: maxar.com



Text Size: A- A+

New Delhi: Pakistani geospatial firm Business Systems International Pvt Ltd (BSI) is no longer listed as a partner on the website of US satellite company Maxar Technologies. This change came hours after ThePrint published an **exclusive report** revealing that Maxar had been

ORDERS FOR PAHALGAM SATELLITE IMAG



* The numbers are indicative

es with pixel
clarity of the

pace Research
e tech startups

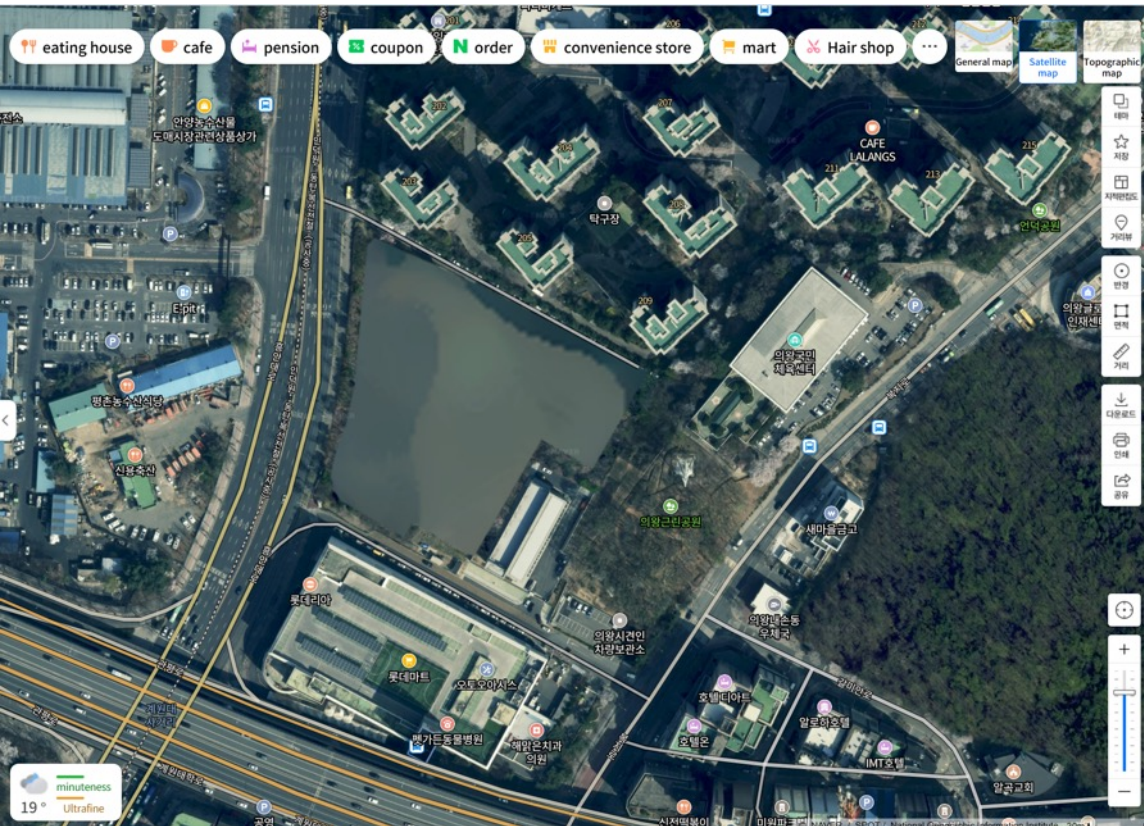
7 2025 with
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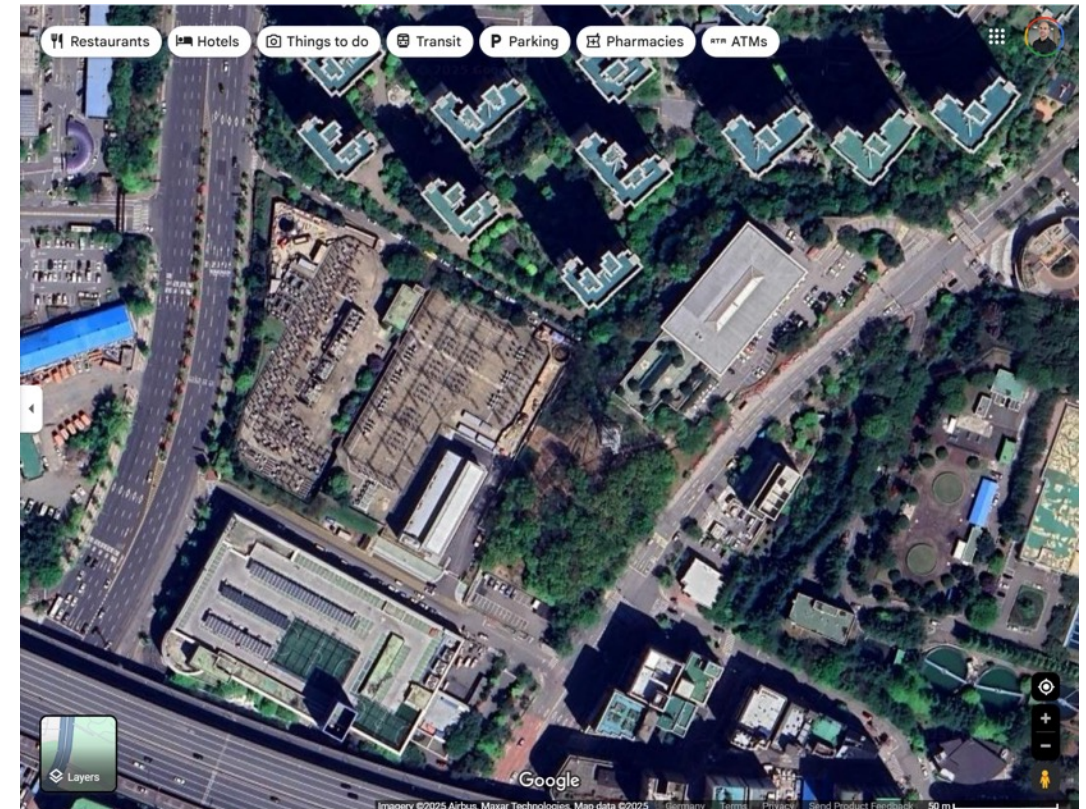
<https://theprint.in/c>

i/

Some Evidence for “Spoofed” Data



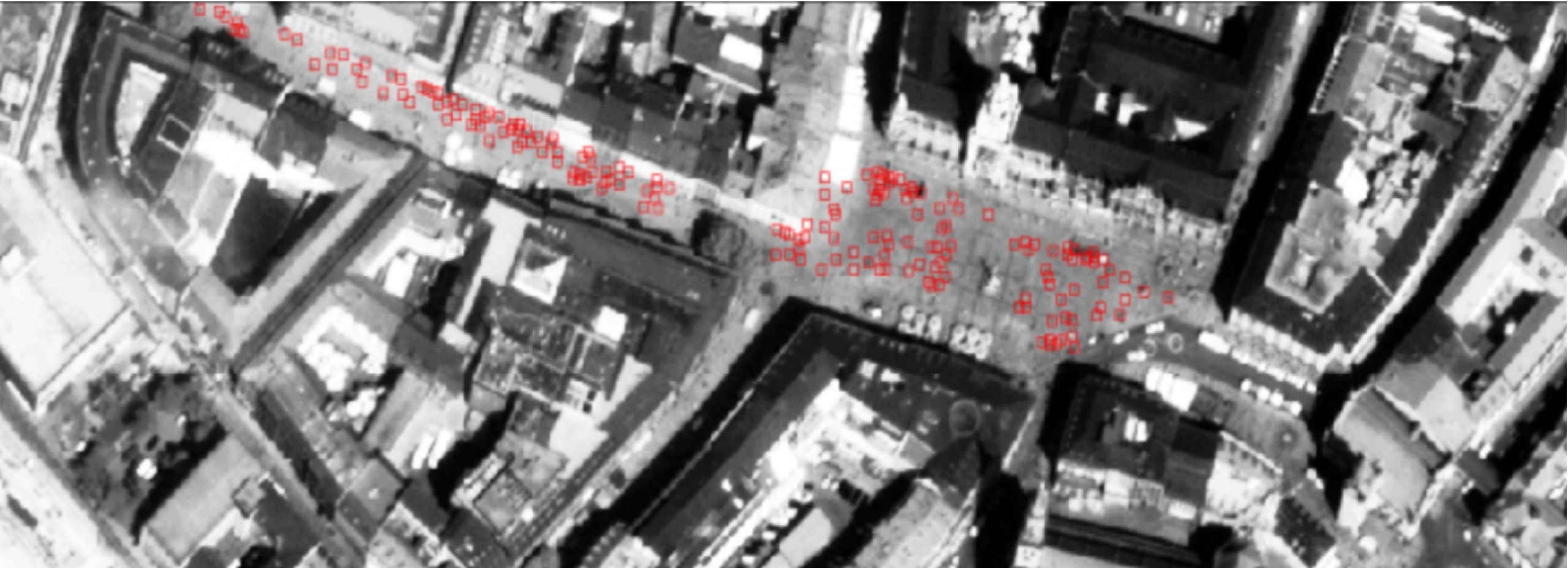
<https://map.naver.com/p/entry/place/36144521?c=17.00,0,0,2,dh>



<https://www.google.com/maps/place/37%C2%B022'50.2%22N+126%C2%B058'22.1%22E/@37.3815569,126.9728885,387m/>

More examples: <https://fas.org/publication/widespread-blurring-of-satellite-images-reveals-secret-facilities/>

Can We Count People?



“Although resolutions of satellite images are still not enough to see each person with sharp contours, we can still notice a slight change of intensity and color components at the place where a person exists.”

Automatic Crowd Analysis from Very High-Resolution Satellite Images

https://www.isprs.org/proceedings/xxxviii/3-w22/pdf/221_XXXVIII-3-W22.pdf

Find Households with Children



These People are “on Camera”



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Commercial

Albedo wins license to sell 10-centimeter imagery

by Debra Werner December 14, 2021



ALBEDO

Albedo plans to operate a constellation of satellites in very low Earth orbit to gather imagery with a resolution of 10 centimeters per pixel. Credit: Albedo

SAN FRANCISCO — Earth observation startup Albedo was granted a National Oceanic and Atmospheric Administration license to sell commercial optical imagery with a resolution of 10 centimeters per pixel.

"We're one step closer to making this mission a reality, capturing 10 centimeter imagery from space and making that data accessible and transparent and easy to purchase," Topher Haddad, Albedo co-founder and CEO, told *SpaceNews*. "Now it's all execution to get to space in the next couple years."

Albedo, a startup based in Austin and Denver, plans to operate a fleet of refrigerator-size satellites to gather electro-optical imagery with 10-centimeter resolution and thermal imagery with four-meter

May 18 - 21

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geoint 2025
SYMPOSIUM

BUILDING A SECURE
TOMORROW TOGETHER

Stellar Dispatch

SpaceWERX Selects Cambrian Works for a \$1.25M Phase II Contract

The NOAA license brings "Albedo one step closer to making aerial quality imagery from space a reality," Albedo said in a Dec. 14 blog post. "10cm satellite imagery will bring another level of transparency and visibility on a global scale."

Specifically, the imagery will help local governments improve urban planning, improve views of forest canopy heights for carbon offset projects and permit accurate roof measurements for insurance and solar companies, according to the blog.

In terms of global transparency, electro-optical imagery of lower resolution revealed possible Uyghur detention camps in China. Albedo imagery will be like "a zoom-in lens" for robust verification, the blog added.

Albedo initially planned to offer two meter resolution thermal imagery but is opting for four meter thermal imagery "due to a more synergistic and low impact compatibility with the optical system," Haddad said.

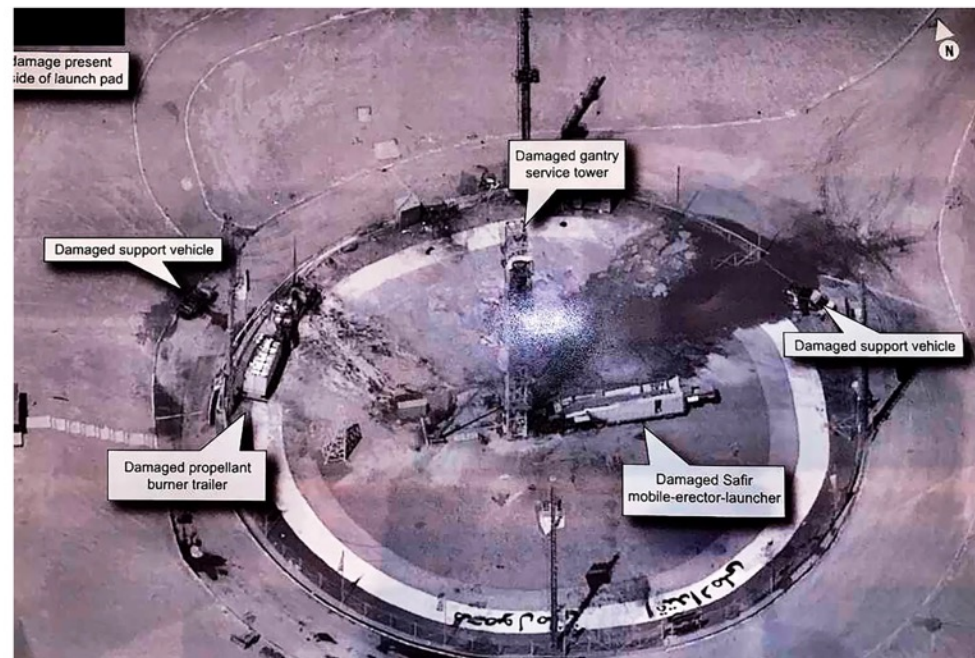
Beyond Commercial Imagery

DANIEL OBERHAUS

SCIENCE SEP 3, 2019 4:45 PM

Trump Tweeted a Sensitive Photo. Internet Sleuths Decoded It

Within hours of the president's post, amateur satellite trackers had hunted down the secret spy satellite that photographed a charred Iranian launchpad.



DONALD J. TRUMP/AP

ON FRIDAY AFTERNOON, President Trump tweeted a photo of a charred launchpad at the Imam Khomeini Space Center in Iran, which had just experienced its third launch failure of the year. Iran's launch accident became public on Thursday after the satellite imaging company Planet released a photo that showed thick black smoke wafting across the launchpad.

Not Just Daytime ...



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Imagine you are a young scholar working in computational social science. You are considering the use of satellite imagery as a data source. What would be reasons for and against this decision? Is the research space already crowded? Is it expected to grow? What are specific application areas where it might be particularly promising to use satellite imagery and where this has not been done before?

Thought for 1m 52s >

Below is a concise "pros, cons, and opportunities" briefing you might give to a graduate seminar in computational social science.

<https://chatgpt.com/share/6825fb38-54e8-800f-83e2-12f3c28c1385>

1 Why satellite imagery is attractive for social-science research

Dimension	Advantages	Typical Data Sources
Spatial-temporal reach	Global coverage (including politically sensitive or data-poor regions) and archives back to the 1970s (Landsat) let you build true panel datasets. New constellations offer daily or even sub-daily refresh. mdpi.com	Landsat, Sentinel-1/2, PlanetScope, Maxar, VIIRS night-lights
Standardisation & comparability	Same sensor → same metric across borders; avoids inconsistent administrative definitions.	Open EO catalogues; Google Earth Engine (GEE)
Cost & access	Medium-resolution imagery (10–30 m) is now free; cloud platforms remove the need for local high-performance hardware.	ESA Copernicus, NASA Earthdata, GEE
Passive & non-reactive	Subjects can't change behaviour in response to being "surveyed," reducing measurement bias.	–
Synergy with ML	Convolutional nets + transfer learning enable automated feature extraction (e.g., roof density → wealth, debris fields → conflict damage).	PyTorch, TensorFlow, fast.ai



2 Drawbacks, caveats, and ethics

1. **Resolution limits & proxy quality** – Many social variables (income, health, gender norms) require indirect indicators; sub-meter imagery is expensive or restricted.
2. **Atmospheric noise & seasonality** – Cloud cover, phenology, and off-nadir angles complicate time-series inference.
3. **Ground-truth dependence** – Models trained on flawed surveys inherit those biases (see fairness critique of poverty maps).
4. **Computation & skill barrier** – Pre-processing (orthorectification, radiometric correction) and GPU training are non-trivial.
5. **Legal & privacy constraints** – EU GDPR treats very-high-resolution (VHR) imagery as potentially personally identifiable; some countries restrict export of sub-1 m pixels. researchgate.net

<https://chatgpt.com/share/6825fb38-54e8-800f-83e2-12f3c28c1385>

3 Is the research space “crowded”?

Indicator	2016	2020	2024
Scopus articles using “satellite imagery” AND “poverty/inequality”	≈20	≈55	> 150
Share of total remote-sensing papers	<1 %	~2 %	~4 %
Share of all social-science papers	negligible	0.2 %	~0.6 %

Bibliometric reviews report exponential growth but also reveal that the **vast majority of remote-sensing work still targets environmental or geophysical questions, not social outcomes.**

Growth drivers

- 7,500 satellites already in orbit, with Earth-observation (EO) platforms the fastest-growing segment.

mdpi.com

researchgate.net
- Commercial providers (Planet Labs, ICEYE, Umbra) releasing academic programs and reduced-cost licenses.
- Generative AI and foundation models for vision tasks reduce annotation costs.

Bottom line: The field is vibrant, not saturated, and capacity is expanding faster than researcher uptake.

TL;DR

Satellite imagery is rapidly becoming a mainstream data source for computational social science because coverage, frequency, and ML tooling have improved dramatically. The research frontier is **far from saturated**; emerging sensors (daily optical, all-weather SAR, forthcoming hyperspectral) and new policy demands (SDGs, climate adaptation, supply-chain transparency) mean the pie is still expanding. Focus on questions where ground data are missing or biased, and where imagery offers a plausible physical proxy—conflict recovery, informal housing, gendered climate risk, and labour-rights oversight are prime candidates.

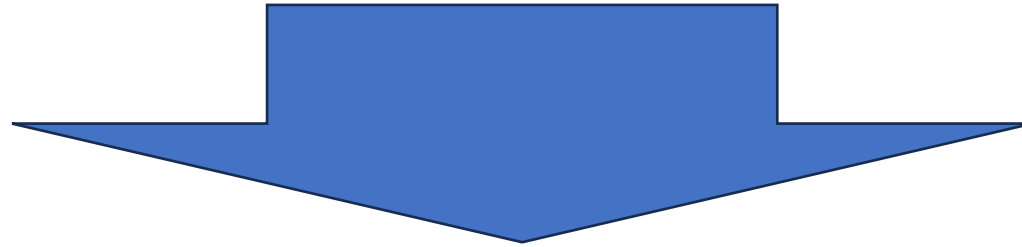
<https://chatgpt.com/share/6825fb38-54e8-800f-83e2-12f3c28c1385>

Takeaway Message

More, better, cheaper
satellite images

+

More, better, easier
computer vision



Possibilities only limited by imagination.
Poorly understood privacy implications.

Thanks!

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